

Trust Region Policy Optimization

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1 Introduction

Most algorithms for policy optimization can be classified into three broad categories:

1. policy iteration methods, which alternate between estimating the value function under the current policy and improving the policy
2. policy gradient methods, which use an estimator of the gradient of the expected return (total reward) obtained from sample trajectories (and which, as we later discuss, have a close connection to policy iteration)
3. derivative-free optimization methods, such as the cross-entropy method (CEM) and covariance matrix adaptation (CMA), which treat the return as a black box function to be optimized in terms of the policy parameters

In this article, we first prove that minimizing a certain surrogate objective function guarantees policy improvement with non-trivial step sizes. Then we make a series of approximations to the theoretically-justified algorithm, yielding a practical algorithm, which we call trust region policy optimization (TRPO).

2 Preliminaries

Consider an infinite-horizon discounted Markov decision process (MDP), defined by the tuple $(\mathcal{S}, \mathcal{A}, P, r, \rho_0, \gamma)$, where $\mathcal{S}$ is a finite set of states, $\mathcal{A}$ is a finite set of actions, $P : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow \mathbb{R}$ is the transition probability distribution, $r : \mathcal{S} \rightarrow \mathbb{R}$ is the reward function, $\rho_0 : \mathcal{S} \rightarrow \mathbb{R}$ is the distribution of the initial state s_0 , and $\gamma \in (0, 1)$ is the discount factor.

“infinite-horizon”: allow infinite interaction

“discounted”: future rewards are worth less than immediate rewards.

The return function at time t is

$$G_t = r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \gamma^3 r_{t+3} + \dots$$

Let π denote a stochastic policy $\pi : \mathcal{S} \times \mathcal{A} \rightarrow [0, 1]$, and let $\eta(\pi)$ denote its expected discounted reward:

$$\eta(\pi) = \mathbb{E}_{s_0, a_0, \dots} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t) \right]$$

where

$$s_0 \sim \rho_0(s_0), a_t \sim \pi(a_t | s_t), s_{t+1} \sim P(s_{t+1} | s_t, a_t)$$

We will use the following standard definitions for **state-action value function** Q_π , the **value function** V_π and the **advantage function** A_π

$$Q_\pi(s_t, a_t) = \mathbb{E}_{s_{t+1}, a_{t+1}, \dots} \left[\sum_{l=0}^{\infty} \gamma^l r(s_{t+l}) \right]$$

$$V_\pi(s_t) = \mathbb{E}_{a_t, s_{t+1}, \dots} \left[\sum_{l=0}^{\infty} \gamma^l r(s_{t+l}) \right]$$

$$A_\pi(s, a) = Q_\pi(s, a) - V_\pi(s) \text{ where}$$

$$a_t \sim \pi(a_t | s_t), s_{t+1} \sim P(s_{t+1} | s_t, a_t) \text{ for } t \geq 0$$

The following useful identity expresses the expected return of another policy $\tilde{\pi}$ in terms of the advantage over π , accumulated over timesteps

$$\eta(\tilde{\pi}) = \eta(\pi) + \mathbb{E}_{s_0, a_0, \dots \sim \tilde{\pi}} \left[\sum_{i=0}^{\infty} \gamma^i A_\pi(s_i, a_i) \right]$$

Let ρ_π be the **(unnormalized) discounted visitation frequencies**:

$$\rho_\pi(s) = P(s_0 = s) + \gamma P(s_1 = s) + \gamma^2 P(s_2 = s) + \dots$$

where $s_0 \sim \rho_0$ and the actions are chosen according to π .

Suppose X and Y are two different random variables. Then

$$\begin{aligned} \mathbb{E}[X + Y] &= \sum_x \sum_y (x + y) p_{X,Y}(x, y) = \sum_x \sum_y x p_{X,Y}(x, y) + \sum_x \sum_y y p_{X,Y}(x, y) \\ &= \sum_x x P(X = x) + \sum_y y P(Y = y) \\ &= \mathbb{E}[X] + \mathbb{E}[Y] \end{aligned}$$

We can rewrite $\eta(\tilde{\pi})$ as

$$\begin{aligned}
\eta(\tilde{\pi}) &= \eta(\pi) + \sum_{t=0}^{\infty} \sum_s P(s_t = s | \tilde{\pi}) \sum_a \tilde{\pi}(a|s) \gamma^t A_{\pi}(s, a) \\
&= \eta(\pi) + \sum_s \sum_{t=0}^{\infty} \gamma^t P(s_t = s | \tilde{\pi}) \sum_a \tilde{\pi}(a|s) A_{\pi}(s, a) \\
&= \eta(\pi) + \sum_s \rho_{\tilde{\pi}}(s) \sum_a \tilde{\pi}(a|s) A_{\pi}(s, a)
\end{aligned} \tag{1}$$

This equation implies that any policy update $\pi \rightarrow \tilde{\pi}$ that has a nonnegative expected advantage at **every** state s , i.e., $\sum_a \tilde{\pi}(a|s) A_{\pi}(s, a) \geq 0$, is guaranteed to increase the policy performance η , or leave it constant in the case that the expected advantage is zero everywhere.

This implies the classic result that the update performed by exact policy iteration, which uses the deterministic policy

$$\tilde{\pi}(s) = \arg \max_a A_{\pi}(s, a),$$

improves the policy if there is at least one state-action pair with a positive advantage value and nonzero state visitation probability, otherwise the algorithm has converged to the optimal policy.

However, in the approximate setting, it will typically be unavoidable, due to estimation and approximation error, that there will be some states s for which the expected advantage is negative, that is, $\sum_a \tilde{\pi}(a|s) A_{\pi}(s, a) < 0$. The complex dependency of $\rho_{\tilde{\pi}}(s)$ on $\tilde{\pi}$ makes Equation (1) difficult to optimize directly. Instead, we introduce the following local approximation to η :

$$L_{\pi}(\tilde{\pi}) = \eta(\pi) + \sum_s \rho_{\pi}(s) \sum_a \tilde{\pi}(a|s) A_{\pi}(s, a)$$

If we have a parameterized policy π_{θ} , where $\pi_{\theta}(a|s)$ is a differentiable function of the parameter vector θ , then L_{π} matches η to first order. That is

$$\begin{aligned}
L_{\pi_{\theta_0}}(\pi_{\theta_0}) &= \eta(\pi_{\theta_0}) \\
\nabla_{\theta} L_{\pi_{\theta_0}}(\pi_{\theta})|_{\theta=\theta_0} &= \nabla_{\theta} \eta(\pi_{\theta})|_{\theta=\theta_0}
\end{aligned} \tag{2}$$

Equation (2) implies that a sufficiently small step $\pi_{\theta_0} \rightarrow \tilde{\pi}$ that improves $L_{\pi_{\theta_{old}}}$ will also improve η , but does not give us any guidance on how big of a step to take

To address this issue, [KL02] proposed a policy updating scheme called **conservative policy iteration**, for which they could provide explicit lower bounds on the improvement of η . To define the conservative policy iteration update, let π_{old} denote the current policy, and let $\pi' = \arg \max_{\pi'} L_{\pi_{old}}(\pi')$. The new policy π_{new} was defined to be the following mixture

$$\pi_{new}(a|s) = (1 - \alpha)\pi_{old}(a|s) + \alpha\pi'(a|s)$$

And we have the following lower bound

$$\eta(\pi_{new}) > L_{\pi_{old}}(\pi_{new}) - \frac{2\epsilon\gamma}{(1 - \gamma)^2}\alpha^2 \quad (3)$$

where $\epsilon = \max_s |\mathbb{E}_{a \sim \pi'(a|s)}[A_{\pi}(s, a)]|$

3 Monotonic Improvement Guarantee for General Stochastic Policies

Our result is that the policy improvement bound in Equation (3) can be extended to general stochastic policies.

The particular distance measure we use is the total variation divergence, which is defined by $D_{TV}(p \parallel q) = \frac{1}{2} \sum_i |p_i - q_i|$ for discrete probability density p, q . Define

$$D_{TV}^{max}(\pi, \tilde{\pi}) = \max_s D_{TV}(\pi(\cdot|s) \parallel \tilde{\pi}(\cdot|s))$$

Theorem 3.1. *Let $\alpha = D_{TV}^{max}(\pi_{old}, \pi_{new})$. Then the following bound holds*

$$\eta(\pi_{new}) \geq L_{\pi_{old}}(\pi_{new}) - \frac{4\epsilon\gamma}{(1 - \gamma^2)}\alpha^2$$

where $\epsilon = \max_{s,a} |A_{\pi_{old}}(s, a)|$

Next, we note the following relationship between the total variation divergence and the KL divergence

$$D_{TV}(p \parallel q)^2 \leq D_{KL}(p \parallel q)$$

$$D_{KL}(p \parallel q) = \sum_x p(x) \log \frac{p(x)}{q(x)}$$

Let $D_{KL}^{max}(\pi, \tilde{\pi}) = \max_s D_{KL}(\pi(\cdot|s) \parallel \tilde{\pi}(\cdot|s))$. The following bound follows from Theorem 3.1

$$\eta(\tilde{\pi}) \geq L_{\pi}(\tilde{\pi}) - CD_{KL}^{max}(\pi, \tilde{\pi}) \quad (4)$$

where $C = \frac{4\epsilon\gamma}{(1-\gamma)^2}$

It follows from Equation (4) that Algorithm 1 is guaranteed to generate a monotonically improving sequence of policies $\eta(\pi_0) \leq \eta(\pi_1) \leq \dots$

Algorithm 1 Policy iteration algorithm guaranteeing non-decreasing expected return η

Initialize π_0 .

for $i = 0, 1, 2, \dots$ until convergence **do**

 Compute all advantage values $A_{\pi_i}(s, a)$.

 Solve the constrained optimization problem

$$\pi_{i+1} = \arg \max_{\pi} [L_{\pi_i}(\pi) - CD_{KL}^{max}(\pi_i, \pi)]$$

$$\text{where } C = 4\epsilon\gamma/(1-\gamma)^2$$

$$\text{and } L_{\pi_i}(\pi) = \eta(\pi_i) + \sum_s \rho_{\pi_i}(s) \sum_a \pi(a|s) A_{\pi_i}(s, a)$$

end for

Figure 1

To see this, let $M_i(\pi) = L_{\pi_i}(\pi) - CD_{KL}^{max}(\pi_i, \pi)$. Then

1. $\eta(\pi_{i+1}) \geq M_i(\pi_{i+1})$ by Equation (4)
2. $\eta(\pi_i) = M_i(\pi_i)$
3. $\eta(\pi_{i+1}) - \eta(\pi_i) \geq M_i(\pi_{i+1}) - M_i(\pi_i)$

4 Optimization of Parameterized Policies

We now describe how to derive a practical algorithm from these theoretical foundations, under finite sample counts and arbitrary parameterizations.

Since we consider parameterized policies $\pi_{\theta}(a|s)$ with parameter vector θ , we will overload our previous notation to use functions of θ rather than π , e.g., $\eta(\theta) := \eta(\pi_{\theta})$, $L_{\theta}(\tilde{\theta}) := L_{\pi_{\theta}}(\pi_{\tilde{\theta}})$, and $D_{KL}(\theta \parallel \tilde{\theta}) := D_{KL}(\pi_{\theta} \parallel \pi_{\tilde{\theta}})$

The preceding section showed that $\eta(\theta) \geq L_{\theta_{old}}(\theta) - CD_{KL}^{max}(\theta_{old}, \theta)$ with equality at $\theta = \theta_{old}$. Thus by performing the following maximization, we are guaranteed to improve the true objective η :

$$\text{maximize}_{\theta} [L_{\theta_{old}}(\theta) - CD_{KL}^{max}(\theta_{old}, \theta)]$$

5 Appendix

Lemma 5.1. *Given two policies $\pi, \hat{\pi}$*

$$\eta(\tilde{\pi}) = \eta(\pi) + \mathbb{E}_{\tau \sim \tilde{\pi}} \left[\sum_{t=0}^{\infty} \gamma^t A_{\pi}(s_t, a_t) \right]$$

This expectation is taken over trajectories $\tau := (s_0, a_0, s_1, a_1, \dots)$, and the notation $\mathbb{E}_{\tau \sim \tilde{\pi}[\dots]}$ indicates actions are sampled from $\tilde{\pi}$ to generate τ

Proof. First note that $A_{\pi}(s, a) = \mathbb{E}_{s' \sim P(s'|s, a)} [r(s) + \gamma V_{\pi}(s') - V_{\pi}(s)]$.

$$\begin{aligned} & \mathbb{E}_{\tau|\tilde{\pi}} \left[\sum_{t=0}^{\infty} \gamma^t A_{\pi}(s_t, a_t) \right] \\ &= \mathbb{E}_{\tau|\tilde{\pi}} \left[\sum_{t=0}^{\infty} \gamma^t (r(s_t) + \gamma V_{\pi}(s_{t+1}) - V_{\pi}(s_t)) \right] \\ &= \mathbb{E}_{\tau|\tilde{\pi}} \left[-V_{\pi}(s_0) + \sum_{t=0}^{\infty} \gamma^t r(s_t) \right] \\ &= -\mathbb{E}_{s_0} [V_{\pi}(s_0)] + \mathbb{E}_{\tau|\tilde{\pi}} \left[\sum_{i=0}^{\infty} \gamma^i r(s_i) \right] \\ &= -\eta(\pi) + \eta(\tilde{\pi}) \end{aligned}$$

□

6 Problems

7 References

References

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